

Lyapunov-Based Target Tracking Control for AIFormula

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System Modeling

The robot system to be studied is depicted in Figure 1. The following notations are used for system description.

$\mathbf{XOY}$, world coordinate in the workspace; $\mathbf{o}(\mathbf{x}, \mathbf{y})$, centre of gravity at the middle of line between left and right wheels; v , linear velocity of the robot; ω , angular velocity of the robot; ϕ , directional angle of robot velocity; $O_t(x_t, y_t)$, position of the target; v_t , velocity of the target; ϕ_t , directional angle of target velocity; ρ , distance between robot and the target; θ , angle of the vector from O_r to O_t .

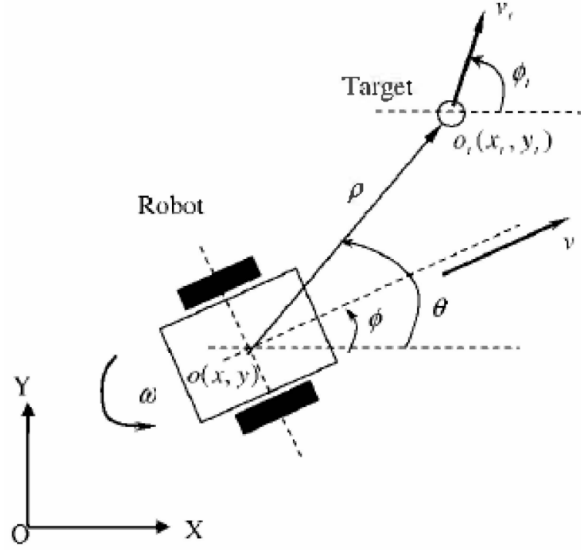


Figure 1: Mobile Robot Chasing a Target

The robot's kinematics is described by

$$\dot{x} = v \cos \theta$$

$$\dot{y} = v \sin \theta$$

$$\dot{\phi} = \omega$$

The linear and angular positions of the robot and the target are related by

$$\rho \cos \theta = x_t - x$$

$$\rho \sin \theta = y_t - y$$

$$\alpha = \theta - \phi$$

$$\beta = \theta - \phi_t$$

Their derivatives with respect to time t are

$$\dot{\rho} = v_t \cos \beta - v \cos \alpha \quad (1)$$

$$\dot{\alpha} = v \frac{\sin \alpha}{\rho} - v_t \frac{\sin \beta}{\rho} - \omega, \rho \neq 0 \quad (2)$$

$$\dot{\beta} = v \frac{\sin \alpha}{\rho} - v_t \frac{\sin \beta}{\rho} - \dot{\phi}_t, \rho \neq 0 \quad (3)$$

This is the kinematic model of the system described by new state variables in polar coordinates: ρ , α and β (where ρ is the linear tracking error and α and β are the angular tracking errors).

Controller Design

The objective of the controller is to make the robot track the moving target. A Lyapunov theory-based controller is proposed, and it computes both the robot's linear and angular velocities to regulate the position and orientation of the robot according to the motion of the moving target. The controller is developed on the basis of the robot's kinematic model.

The following candidate Lyapunov function is chosen for controller design

$$V = V_1 + V_2 \quad (4)$$

$$V_1 = \frac{1}{2} \rho^2 \quad (5)$$

$$V_2 = \frac{1 - \cos \alpha}{k_1} + \frac{1 - \cos \beta}{k_2} \quad (6)$$

where, V_1 and V_2 are designed for the linear and angular tracking errors respectively.

Differentiating V_1 with respect to time t and considering (1),

$$\dot{V}_1 = \rho (v_t \cos \beta - v \cos \alpha) \quad (7)$$

Choice of v to make $\dot{V}_1$ non-positive is,

$$v = \frac{v_t \cos \beta}{\cos \alpha} + \lambda_v \rho \cos \alpha \quad (8)$$

where $\lambda_v > 0$ is a control parameter. Substituting this v into,

$$\dot{V}_1 = -\lambda_v \rho^2 \cos^2 \alpha \leq 0 \quad (9)$$

thus v makes $\dot{V}_1$ non-positive and ρ convergent. But v is in proportional relation with ρ , $v_t \cos \beta$ and $\cos \alpha$ which have large magnitudes especially at the initial state of the system.

This becomes worse when designing ω to make V_2 non-negative.

So v is modified as

$$v = (v_t \cos \beta + A_v \rho) \cos \alpha, \quad (10)$$

Substituting into (7) and (1)

$$\dot{V}_1 = -\lambda_v \rho^2 \cos^2 \alpha + v_t \rho \sin^2 \alpha \cos \beta \quad (11)$$

$$\dot{\rho} = -\lambda_v \rho \cos^2 \alpha + v_t \sin^2 \alpha \cos \beta \quad (12)$$

From (II), it is clear that $\dot{V}_2$ will not be non-positive until $\alpha \approx 0$; the new v in (10) is designed so as to reduce the contribution from the target velocity v_t , by multiplying with $\cos \alpha$ and $\cos \beta$ (either of which is bounded by the magnitude). Contribution of ρ is also reduced by multiplying with $\cos \alpha$.

For finding the controller input ω accordingly, differentiating V_2 with respect to time t and considering (2), (3) and (10)

$$V_2 = \left(\frac{\sin \alpha}{k_1 \rho} + \frac{\sin \beta}{k_2 \rho} \right) \left\{ \left(\frac{\sin 2\alpha}{2} \cos \beta - \sin \beta \right) v_t - \omega \frac{\sin \alpha}{k_1} - \frac{\dot{\phi}_t \sin \beta}{k_2} + \frac{\sin 2\alpha}{2} \lambda_v \left(\frac{\sin \alpha}{k_1} + \frac{\sin \beta}{k_2} \right) \right\} \quad (13)$$

Letting

$$\omega = \lambda_a \sin \alpha + \left(\frac{\sin \alpha}{k_1 \rho} + \frac{\sin \beta}{k_2 \rho} \right) \left[\left(\frac{k_1 \sin 2\alpha}{2 \sin \alpha} \cos \beta - \frac{k_1 \sin \beta}{\sin \alpha} \right) v_t - \frac{\dot{\phi}_t k_1 \sin \beta}{k_2 \sin \alpha} + \frac{k_1 \sin 2\alpha}{2 \sin \alpha} \lambda_v \left(\frac{\sin \alpha}{k_1 \rho} + \frac{\sin \beta}{k_2 \rho} \right) \right] \quad (14)$$

Substituting in (13),

$$V_2 = -\frac{\lambda_a \sin^2 \alpha}{k_1} \quad (15)$$

This shows that V_2 is non-increasing. Given $V_2 > 0$ as defined, V_2 converges to a non-negative limit asymptotically. Similar is the case for V_1 also. As a result, the state variables ρ , α and β are all bounded.

Thus with the designed controller, $\alpha \rightarrow 0, \rho \rightarrow 0, \beta \rightarrow 0$, can be achieved when the robot tracks a moving target which moves at a velocity v_t with direction θ_t .